Question Bank with Answers — AI Quest

72 questions on Units I & II of Artificial Intelligence (DSC401-6)

Name	Manish Nayak
Register No.	2441442
Class	6 BSc DS
Department	Data Science & Statistics
University	Christ University, Bangalore

This document lists all 72 questions of the AI Quest quiz game with their correct answers and explanations. Questions are grouped by topic in the order of the syllabus. Tags: [RW] = real-world application/scenario question, [S] = search/state-space problem. In the game, MCQ options are shuffled and Match-the-Following right-hand items are shuffled; they are listed here in their stored order.

Shared search graph used by several Problem-solving questions: edges (cost) S→A(2), S→B(3), A→C(4), A→D(3), B→D(2), B→E(6), C→G(3), D→G(5), E→G(1); heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0; start S, goal G; ties broken alphabetically.

1. Foundations of AI

Q01 *MCQ · Easy*

A chatbot is engineered with one goal: to be indistinguishable from a human in conversation. Which of Russell & Norvig's four approaches to AI does this design follow?

(A) Thinking humanly
(B) Acting humanly
(C) Thinking rationally
(D) Acting rationally

Answer: (B) Acting humanly

Explanation: Building a system to pass as human in behaviour is the 'acting humanly' (Turing Test) approach. 'Acting rationally' would instead optimise for the best outcome, even if no human would act that way.

Q02 *Assertion-Reason · Medium*

Assertion (A): The 'laws of thought' (thinking rationally) approach is hard to apply directly to real-world problems.

Reason (R): Informal, uncertain knowledge is difficult to express in formal logic, and solving a problem 'in principle' differs greatly from solving it in practice.

(A) Both A and R are true, and R is the correct explanation of A
(B) Both A and R are true, but R is NOT the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Both statements are true and R is precisely why the logicist tradition struggles: uncertainty doesn't fit neatly into logic, and even well-posed logical problems can exhaust computational resources.

Q03 *Fill in the Blank · Easy*

The term 'Artificial Intelligence' was coined at the _____ workshop in 1956, organised by John McCarthy.

Answer: Dartmouth (also accepted: dartmouth conference, dartmouth workshop, dartmouth college)

Explanation: The 1956 Dartmouth Summer Research Project (McCarthy, Minsky, Rochester, Shannon) is considered the birth of AI as a field.

Q04 *True/False · Medium*

Under the 'thinking humanly' (cognitive modelling) approach, a program is validated only by whether it produces the correct output.

Answer: False

Explanation: Cognitive modelling requires the program's reasoning steps to match human ones (validated against psychological experiments or brain imaging). Correct output alone is the standard of acting/thinking rationally.

2. Applications of AI

Q05 *Scenario-based · Easy [RW]*

A hospital deploys a model that reads chest X-rays and flags likely pneumonia; a radiologist confirms every flagged case. Which statement best describes this system?

(A) Artificial General Intelligence (AGI)

(B) Narrow (weak) AI used as human-in-the-loop decision support

(C) A fully autonomous agent with no human oversight

(D) A production system with no learning component

Answer: (B) Narrow (weak) AI used as human-in-the-loop decision support

Explanation: It solves one specific task, has no general intelligence, and a human makes the final decision — classic narrow AI in a decision-support role.

Q06 *Match the Following · Easy [RW]*

Match each real-world application with the AI area it mainly relies on.

Column A	Column B (correct match)
Google Maps route planning	Search & planning
Gmail spam filter	Machine learning (classification)
Alexa understanding a spoken request	Speech & natural language processing
Self-driving car detecting lane markings	Computer vision

Answer: as matched in the table above

Explanation: Route planning is search/planning; spam filtering is supervised classification (ML); voice assistants depend on speech & NLP; lane detection is computer vision.

Q07 *Scenario-based · Medium [RW]*

A UPI fraud-detection agent is evaluated only on accuracy. Just 0.1% of transactions are fraudulent. What is the main flaw in this performance measure?

(A) Accuracy cannot be computed for real-time systems

(B) An agent that labels every transaction 'legitimate' scores 99.9% accuracy while catching zero fraud

(C) Accuracy over-penalises false positives

(D) Nothing — accuracy is the ideal measure here

Answer: (B) An agent that labels every transaction 'legitimate' scores 99.9% accuracy while catching zero fraud

Explanation: With heavy class imbalance, accuracy rewards doing nothing. A good performance measure should reflect what we actually want: e.g. recall on fraud and the real cost of a missed fraud vs a blocked genuine payment.

3. Turing Test

Q08 *MCQ · Easy*

Why does the standard Turing Test have the interrogator communicate only through a text (teletype) channel?

(A) To make the test faster to run

(B) So that physical appearance or voice cannot reveal which is the machine

(C) Because 1950s computers could not process speech

(D) To test the machine's typing speed

Answer: (B) So that physical appearance or voice cannot reveal which is the machine

Explanation: Turing deliberately avoided direct physical interaction because physical simulation of a person is unnecessary for intelligence.

Q09 *Assertion-Reason · Hard*

Assertion (A): A machine that passes the Turing Test has been proven to genuinely understand language.

Reason (R): Searle's Chinese Room argument claims that manipulating symbols by rules is not sufficient for understanding.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (D) A is false, but R is true

Explanation: A is false — the Turing Test is a behavioural test; passing it shows indistinguishable behaviour, not proven understanding. R is a true description of Searle's argument (which is exactly an objection to A).

Q10 *True/False · Medium*

The 'Total Turing Test' additionally requires the machine to have computer vision and robotics capabilities.

Answer: True

Explanation: The Total Turing Test includes a video signal (perception → computer vision) and the ability to pass physical objects (manipulation → robotics), on top of NLP, knowledge representation, reasoning and learning.

Q11 *Scenario-based · Medium [RW]*

When you log in to IRCTC you must solve a CAPTCHA. Why is CAPTCHA often called a 'reverse Turing Test'?

(A) A human judges whether the website is a bot

(B) A computer acts as the judge, deciding whether the user is a human

(C) It is solved in reverse order of the Turing Test questions

(D) It tests whether a machine can create images

Answer: (B) A computer acts as the judge, deciding whether the user is a human

Explanation: In the Turing Test a human judges whether the other party is a machine. In CAPTCHA, the roles are reversed: a machine judges whether you are human.

4. Rational Agents

Q12 *MCQ · Medium*

Which statement correctly defines a rational agent?

(A) It always achieves the best possible actual outcome

(B) It selects the action expected to maximise its performance measure, given its percept sequence and built-in knowledge

(C) It thinks the same way a human expert would

(D) It never takes an action that leads to failure

Answer: (B) It selects the action expected to maximise its performance measure, given its percept sequence and built-in knowledge

Explanation: Rationality is about expected performance given what the agent has perceived and knows — not about perfect outcomes.

Q13 *True/False · Easy*

A rational agent must be omniscient, i.e. it must know the actual outcome of its actions in advance.

Answer: False

Explanation: Rationality ≠ omniscience. Rationality maximises expected performance; an agent crossing a road can be rational and still be hit by a falling aircraft door it could not have perceived.

Q14 *Fill in the Blank · Medium*

An agent that relies on its designer's prior knowledge rather than on its own percepts and learning is said to lack _____.

Answer: autonomy (also accepted: autonomous)

Explanation: A rational agent should be autonomous — it should learn what it can to compensate for partial or incorrect prior knowledge.

5. Good Behaviour

Q15 *Scenario-based · Medium*

A vacuum-cleaner agent is rewarded +1 for every 'Suck' action it performs. What is the most likely unintended behaviour?

(A) It stops once all squares are clean

(B) It cleans, dumps the dirt back, and cleans again forever

(C) It refuses to move

(D) It becomes an omniscient agent

Answer: (B) It cleans, dumps the dirt back, and cleans again forever

Explanation: Rewarding the action rather than the outcome invites gaming. Russell & Norvig's rule: design performance measures according to what you want in the environment (e.g. +1 per clean square per time step), not according to how you think the agent should behave.

Q16 *Assertion-Reason · Medium*

Assertion (A): A rational agent should look both ways before crossing a road, even though looking takes time.

Reason (R): Rationality maximises expected performance, and gathering information can raise the expected value of later actions.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Information gathering (exploration) is part of rational behaviour precisely because it improves expected outcomes — R explains A.

6. Intelligent Agents

Q17 *Match the Following · Medium*

Match each agent type with the example that best fits it.

Column A	Column B (correct match)
Simple reflex agent	Thermostat switching heater on below 20°C
Model-based reflex agent	Driver tracking a car now in the blind spot
Goal-based agent	Robot planning a path to reach the charging dock
Utility-based agent	Ride app choosing the route balancing time, fare & comfort

Answer: as matched in the table above

Explanation: A thermostat maps percept → action directly (simple reflex). Tracking a car you can no longer see needs an internal model of the world. Navigation to a destination is goal-based. Trading off time, fare and comfort requires a utility function.

Q18 *Scenario-based · Easy [RW]*

A smart traffic signal switches to green for the lane whose current queue sensor reading is longest, using only IF-THEN rules on the present reading. What type of agent is it?

(A) Simple reflex agent

(B) Model-based reflex agent

(C) Goal-based agent

(D) Learning agent

Answer: (A) Simple reflex agent

Explanation: It acts on the current percept only via condition-action rules — no memory of past queues, no explicit goal or utility.

Q19 *MCQ · Medium*

In a learning agent, which component suggests exploratory actions that may lead to new and informative experiences?

(A) Performance element

(B) Critic

(C) Learning element

(D) Problem generator

Answer: (D) Problem generator

Explanation: Performance element chooses actions; critic gives feedback against a performance standard; learning element improves the performance element; problem generator proposes exploratory actions.

Q20 *Assertion-Reason · Easy*

Assertion (A): Simple reflex agents work reliably only in fully observable environments.

Reason (R): They select actions based only on the current percept and ignore percept history.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Because they have no memory, anything not visible in the current percept is lost — so they fail (e.g. loop forever) under partial observability.

7. Agents and Environments

Q21 *Fill in the Blank · Easy*

Mathematically, an agent function maps any given percept _____ to an action.

Answer: sequence (also accepted: percept sequence, history, percept history)

Explanation: The agent function f: $P^* \rightarrow A$ takes the complete history of everything the agent has perceived.

Q22 *True/False · Medium*

The agent function and the agent program are the same thing.

Answer: False

Explanation: The agent function is an abstract mathematical mapping; the agent program is its concrete implementation running on a physical architecture. Agent = architecture + program.

Q23 *MCQ · Easy*

Which equation describes an agent according to Russell & Norvig?

(A) Agent = Sensors + Goals

(B) Agent = Architecture + Program

(C) Agent = Environment + Percepts

(D) Agent = Knowledge + Search

Answer: (B) Agent = Architecture + Program

Explanation: The architecture is the computing device with sensors and actuators; the program implements the agent function on it.

8. Nature of Environments

Q24 *Match the Following · Hard*

Match each task environment with the property that best distinguishes it.

Column A	Column B (correct match)
Crossword puzzle	Deterministic and static
Poker	Partially observable, multi-agent
Taxi driving	Dynamic and continuous
Inspecting parts on a conveyor belt	Episodic

Answer: as matched in the table above

Explanation: Crossword: deterministic and static. Poker: hidden cards and an opponent → partially observable, multi-agent. Taxi driving: world changes while you think and positions/speeds are continuous. Defect inspection on an assembly line: each image is judged independently → episodic.

Q25 *Scenario-based · Easy [RW]*

PEAS for a food-delivery routing agent (Swiggy/Zomato). Which of these is a SENSOR?

(A) Live GPS location of the rider

(B) Pushing the route to the rider's app

(C) Minimising average delivery time

(D) City roads and traffic

Answer: (A) Live GPS location of the rider

Explanation: GPS feed is a percept source (sensor). Sending the route to the rider app is an actuator, minimising delivery time is the performance measure, and roads/traffic are the environment.

Q26 *MCQ · Medium*

Chess played with a clock is best classified as:

(A) Static

(B) Dynamic

(C) Semi-dynamic

(D) Continuous

Answer: (C) Semi-dynamic

Explanation: The board doesn't change while you think, but your performance score (remaining time) does — that is semi-dynamic.

Q27 *True/False · Easy*

A medical-diagnosis agent operates in a partially observable and stochastic environment.

Answer: True

Explanation: The patient's internal state can't be observed directly (partially observable) and disease progression / treatment response are uncertain (stochastic).

9. Problems and Problem Spaces

Q28 *MCQ · Easy*

Which of the following is NOT part of a formal problem definition for search?

(A) Initial state

(B) Goal test

(C) Path cost

(D) Heuristic function

Answer: (D) Heuristic function

Explanation: A problem is defined by initial state, actions, transition model, goal test and path cost. A heuristic belongs to a particular search strategy, not to the problem itself.

Q29 *Problem-solving · Hard [S]*

Water-jug problem: a 4-litre jug and a 3-litre jug, both empty, with an unlimited tap. Operators: fill a jug, empty a jug, pour one into the other until one is full/empty. What is the MINIMUM number of operations to get exactly 2 litres in the 4-litre jug?

Answer: 6 (also accepted: six)

Explanation: BFS over states (x, y) gives a shortest path of 6: $(0,0) \to (4,0) \to (1,3) \to (1,0) \to (0,1) \to (4,1) \to (2,3)$. (A different 6-step route via filling the 3-litre jug first also exists.)

Q30 *Fill in the Blank · Easy*

The set of all states reachable from the initial state by any sequence of actions is called the _____.

Answer: state space (also accepted: statespace, problem space, search space)

Explanation: The state space forms a directed graph whose nodes are states and whose edges are actions.

10. State Space Search

Q31 *Problem-solving · Medium [S]*

How many distinct states of the 8-puzzle are reachable from any given start configuration?

(A) 362,880

(B) 181,440

(C) 40,320

(D) 9

Answer: (B) 181,440

Explanation: There are 9! = 362,880 arrangements, but they split into two disjoint halves by parity; only 9!/2 = 181,440 are reachable from any start.

Q32 *Problem-solving · Medium [S]*

Missionaries & Cannibals: state = (M, C, B) on the LEFT bank, start (3, 3, 1). Missionaries must never be outnumbered on either bank where missionaries are present. Which state is INVALID?

(A) (3, 1, 0)

(B) (2, 2, 1)

(C) (1, 2, 1)

(D) (0, 3, 0)

Answer: (C) (1, 2, 1)

Explanation: (1, 2, 1): the left bank has 1 missionary and 2 cannibals → missionary is outnumbered. The others are safe on both banks, e.g. (0,3,0) leaves no missionary on the left and 3 M vs 0 C on the right.

Q33 *True/False · Easy*

Tree search on a state space containing cycles can loop forever unless repeated states are detected.

Answer: True

Explanation: Without an explored set (graph search) or cycle check, a path like $A \to B \to A \to B \ldots$ can be generated indefinitely.

11. Production Systems

Q34 *MCQ · Easy*

In a production system, when several rules match the current working memory, which component decides which one fires?

(A) The rule base

(B) Working memory

(C) Control strategy (conflict resolution)

(D) The goal test

Answer: (C) Control strategy (conflict resolution)

Explanation: A production system = set of rules + database/working memory + control strategy + rule applier. The control strategy resolves conflicts.

Q35 *Match the Following · Hard*

Match each problem with its production-system class (Rich & Knight).

Column A	Column B (correct match)
Theorem proving	Monotonic, partially commutative
Robot navigation	Non-monotonic, partially commutative
Chemical synthesis	Monotonic, not partially commutative
Playing bridge	Non-monotonic, not partially commutative

Answer: as matched in the table above

Explanation: Theorem proving: monotonic & partially commutative. Robot navigation: non-monotonic but partially commutative. Chemical synthesis: monotonic but not partially commutative (order of reactions matters). Bridge (card game): non-monotonic and not partially commutative.

Q36 *Assertion-Reason · Medium*

Assertion (A): A good control strategy for a production system must cause motion.

Reason (R): A strategy that, for example, always fires the first applicable rule may keep producing the same state and never reach a solution.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Rich & Knight's two requirements: the control strategy must cause motion, and it must be systematic. R illustrates exactly why motion is required.

12. Problem Characteristics

Q37 *Match the Following · Medium*

Match each problem with how its solution steps can be treated.

Column A	Column B (correct match)
Theorem proving	Ignorable
8-puzzle	Recoverable
Chess	Irrecoverable

Answer: as matched in the table above

Explanation: Theorem proving: a useless lemma can simply be ignored. 8-puzzle: a wrong move can be undone (recoverable). Chess: a move once made cannot be taken back (irrecoverable).

Q38 *MCQ · Medium*

Which of the following problems is DECOMPOSABLE into independent sub-problems?

(A) Symbolic integration of $\int(x^2 + 3x + \sin^2 x \cdot \cos^2 x)\, dx$

(B) Blocks-world stacking with interacting sub-goals

(C) Playing a game of chess

(D) Solving the 8-puzzle

Answer: (A) Symbolic integration of $\int(x^2 + 3x + \sin^2 x \cdot \cos^2 x)\, dx$

Explanation: The integral splits into three independent integrals solved separately and summed. In the blocks world, sub-goals interact, so it is not simply decomposable.

Q39 *True/False · Medium*

The Travelling Salesman Problem is an 'any-path' problem: any tour that visits every city is an acceptable solution.

Answer: False

Explanation: TSP is a 'best-path' problem — we need the shortest tour, so a good solution is relative, not absolute. Any-path search is not enough.

Q40 *Scenario-based · Hard [RW]*

A banking chatbot must interpret the query 'Block the card I used at the mall yesterday'. In Rich & Knight's terms, what is the solution to this problem?

(A) A path — the sequence of reasoning steps

(B) A state — the correct interpretation — not the reasoning path used

(C) Both — the path must be shown to the user

(D) Neither — it is a decomposable problem

Answer: (B) A state — the correct interpretation — not the reasoning path used

Explanation: Natural-language interpretation is a 'solution is a state' problem: what matters is the final interpretation, not the record of how it was reached (unlike, say, the water-jug problem where the path is the answer).

13. Issues in Search Program Design

Q41 *MCQ · Hard*

When is backward search (from goal to start) generally preferable to forward search?

(A) Always — backward search is optimal

(B) When there are fewer goal states than start states and the backward branching factor is smaller

(C) When the goal state is unknown

(D) Only when the state space is infinite

Answer: (B) When there are fewer goal states than start states and the backward branching factor is smaller

Explanation: The direction choice depends on the relative number of start vs goal states and the branching factor in each direction (e.g. theorem proving or diagnosis often reason backward).

Q42 *True/False · Medium*

Using graph search instead of tree search avoids re-expanding repeated states, at the cost of extra memory for the explored set.

Answer: True

Explanation: Representing the space as a graph (with a closed/explored list) trades memory for not duplicating work — one of the design issues Rich & Knight list (tree vs graph).

14. Breadth-First Search

Q43 *Problem-solving · Medium [S]*

Graph: $S \to A(2)$, $S \to B(3)$, $A \to C(4)$, $A \to D(3)$, $B \to D(2)$, $B \to E(6)$, $C \to G(3)$, $D \to G(5)$, $E \to G(1)$. Heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0. Start = S, Goal = G. Ties / children are taken in alphabetical order.

Ignoring costs, in what order does BFS EXPAND nodes until it expands G (goal test on expansion, repeated states skipped)?

(A) S, A, C, G

(B) S, A, B, C, D, E, G

(C) S, B, E, G

(D) S, A, B, D, C, E, G

Answer: (B) S, A, B, C, D, E, G

Explanation: Level 0: S. Level 1: A, B. Level 2: C, D (from A), E (from B; D already queued). Level 3: G. BFS returns $S \to A \to C \to G$.

Q44 *Assertion-Reason · Medium*

Assertion (A): BFS is optimal when all step costs are equal.

Reason (R): BFS always expands the shallowest unexpanded node first.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Expanding shallowest first means the first goal found has the fewest steps; when every step costs the same, fewest steps = lowest cost.

Q45 *Scenario-based · Easy [RW]*

LinkedIn labels people as 1st, 2nd or 3rd-degree connections. Which search strategy naturally computes these degrees?

(A) Depth-First Search

(B) Breadth-First Search

(C) Hill Climbing

(D) Greedy Best-First Search

Answer: (B) Breadth-First Search

Explanation: BFS from your profile visits all people 1 hop away, then 2 hops, then 3 — exactly the 'degree' of connection.

Q46 *MCQ · Medium*

With branching factor b and shallowest goal at depth d, what is the space complexity of BFS?

(A) O(b·d)

(B) O(d)

(C) O(b^d)

(D) O(d^b)

Answer: (C) O(b^d)

Explanation: BFS must keep the whole frontier in memory, which is O(b^d) — memory, not time, is usually its real bottleneck.

15. Depth-First Search

Q47 *Problem-solving · Easy [S]*

Graph: $S \to A(2)$, $S \to B(3)$, $A \to C(4)$, $A \to D(3)$, $B \to D(2)$, $B \to E(6)$, $C \to G(3)$, $D \to G(5)$, $E \to G(1)$. Heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0. Start = S, Goal = G. Ties / children are taken in alphabetical order.

How many nodes does DFS expand (including S and G) before reaching the goal?

(A) 3

(B) 4

(C) 6

(D) 7

Answer: (B) 4

Explanation: DFS goes S → A → C → G, expanding 4 nodes, and returns $S \to A \to C \to G$ (cost 9). It got lucky here — on other graphs DFS can return a long, costly path.

Q48 *True/False · Easy*

DFS is complete even on state spaces of infinite depth.

Answer: False

Explanation: DFS can go down an infinite branch forever. It is complete only on finite spaces (with repeated-state checking).

Q49 *Fill in the Blank · Easy*

DFS stores its frontier in a _____ (LIFO) data structure.

Answer: stack (also accepted: lifo stack)

Explanation: BFS uses a FIFO queue; DFS uses a LIFO stack (or recursion, which uses the call stack).

16. Iterative Deepening

Q50 *Problem-solving · Hard [S]*

Graph: S→A(2), S→B(3), A→C(4), A→D(3), B→D(2), B→E(6), C→G(3), D→G(5), E→G(1). Heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0. Start = S, Goal = G. Ties / children are taken in alphabetical order.

Run Iterative Deepening DFS with limits 0, 1, 2, … (no repeated-state checking; stop as soon as G is expanded). What is the TOTAL number of node expansions across all iterations?

(A) 7

(B) 12

(C) 15

(D) 19

Answer: (C) 15

Explanation: Limit 0: S (1). Limit 1: S, A, B (3). Limit 2: S, A, C, D, B, D, E (7 — D appears twice, via A and via B). Limit 3: S, A, C, G (4). Total = 1 + 3 + 7 + 4 = 15.

Q51 *Assertion-Reason · Hard*

Assertion (A): Iterative deepening regenerates upper levels many times, yet its time overhead over BFS is small.

Reason (R): In a tree with a constant branching factor, most nodes lie at the deepest level.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Total generated ≈ d·b + (d−1)·b² + … + 1·b^d = O(b^d). Because the bottom level dominates, repeating the shallow levels adds only a small constant factor (~b/(b−1)).

Q52 *Scenario-based · Medium [RW]*

Chess engines such as Stockfish use iterative deepening under a game clock. What is the main practical benefit?

(A) It uses more memory than BFS so it is more accurate

(B) When time runs out, the engine can play the best move from the last fully completed depth

(C) It guarantees a checkmate is found

(D) It removes the need for an evaluation function

Answer: (B) When time runs out, the engine can play the best move from the last fully completed depth

Explanation: Iterative deepening is an 'anytime' strategy: it always has an answer from the deepest finished iteration, and earlier iterations also give good move ordering for the next.

17. Hill Climbing

Q53 *Problem-solving · Medium [S]*

Objective values along a 1-D landscape (index 0–9): [3, 5, 8, 6, 4, 7, 9, 12, 10, 2]. Neighbours are index ± 1. Steepest-ascent hill climbing (maximising) starts at index 0. What value does it stop at?

(A) 8

(B) 9

(C) 12

(D) 10

Answer: (A) 8

Explanation: 0 (3) → 1 (5) → 2 (8). Both neighbours of index 2 (5 and 6) are lower, so it halts at a LOCAL maximum of 8 — the global maximum 12 (index 7) is never reached.

Q54 *Match the Following · Medium*

Match each hill-climbing failure with the remedy suggested by Rich & Knight.

Column A	Column B (correct match)
Local maximum	Backtrack and try a different direction
Plateau	Make a big jump to a new part of the space
Ridge	Apply two or more rules before testing

Answer: as matched in the table above

Explanation: Local maximum → backtrack to an earlier node and try another direction. Plateau → make a big jump to a new region. Ridge → apply two or more rules (move in several directions) before testing.

Q55 *MCQ · Hard*

In simulated annealing a worse move ($\Delta E < 0$) is accepted with probability $e^{(\Delta E/T)}$. As the temperature $T \to 0$, the algorithm behaves like:

(A) A random walk

(B) Hill climbing (only uphill moves)

(C) Breadth-first search

(D) A* search

Answer: (B) Hill climbing (only uphill moves)

Explanation: As $T \to 0$, $e^{(\Delta E/T)} \to 0$ for any $\Delta E < 0$, so downhill moves are never accepted — it degenerates into (stochastic) hill climbing. At high T it behaves like a random walk.

Q56 *Scenario-based · Medium [RW]*

A quick-commerce company uses hill climbing to choose a dark-store location that minimises average delivery distance. Different starting points give different final locations. What is the most practical fix?

(A) Use a smaller step size only

(B) Random-restart hill climbing: run from many random starts and keep the best

(C) Switch to DFS

(D) Stop after the first improving move

Answer: (B) Random-restart hill climbing: run from many random starts and keep the best

Explanation: Different results from different starts = local optima. Random restarts explore several basins and keep the best one, approaching the global optimum with probability → 1 as restarts increase.

18. Heuristic Search

Q57 *Problem-solving · Medium [S]*

8-puzzle. Start: 2 8 3 / 1 6 4 / 7 _ 5. Goal: 1 2 3 / 8 _ 4 / 7 6 5. h1 = number of misplaced tiles, h2 = sum of Manhattan distances (blank not counted). What are h1 and h2 for the start state?

(A) h1 = 4, h2 = 5

(B) h1 = 5, h2 = 4

(C) h1 = 3, h2 = 5

(D) h1 = 4, h2 = 6

Answer: (A) h1 = 4, h2 = 5

Explanation: Misplaced: 2, 8, 1, 6 → h1 = 4. Manhattan: 2→1, 8→2, 1→1, 6→1, others 0 → h2 = 5. (The true optimal solution is 5 moves, so h2 is exact here.)

Q58 *Assertion-Reason · Hard*

Assertion (A): For A* on the 8-puzzle, Manhattan distance (h2) is a better heuristic than misplaced tiles (h1).

Reason (R): Both are admissible, and $h2(n) \geq h1(n)$ for every state n (h2 dominates h1).

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: A dominating admissible heuristic is closer to the true cost, so A* with h2 never expands more nodes than with h1 (apart from ties).

Q59 *True/False · Medium*

If a heuristic overestimates the true cost to the goal, A* may return a non-optimal solution.

Answer: True

Explanation: Admissibility (never overestimating) is what guarantees A* (tree search) optimality. An overestimate can make the optimal path look worse than a sub-optimal one.

19. Best First Search

Q60 *Problem-solving · Medium [S]*

Graph: S→A(2), S→B(3), A→C(4), A→D(3), B→D(2), B→E(6), C→G(3), D→G(5), E→G(1). Heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0. Start = S, Goal = G. Ties / children are taken in alphabetical order.

Which path does GREEDY best-first search (f = h) return, and what is its cost?

(A) S → A → C → G, cost 9

(B) S → B → E → G, cost 10

(C) S → B → D → G, cost 10

(D) S → A → D → G, cost 10

Answer: (B) S → B → E → G, cost 10

Explanation: From S, B (h=4) beats A (h=5). From B, E (h=1) beats D (h=4). E → G. Path cost 3 + 6 + 1 = 10 — not optimal (S→A→C→G costs 9), showing greedy search ignores g(n).

Q61 *Problem-solving · Hard [S]*

Graph: S→A(2), S→B(3), A→C(4), A→D(3), B→D(2), B→E(6), C→G(3), D→G(5), E→G(1). Heuristic h: S=7, A=5, B=4, C=2, D=4, E=1, G=0. Start = S, Goal = G. Ties / children are taken in alphabetical order.

Which path does A* search (f = g + h) return, and what is its cost?

(A) S → B → E → G, cost 10

(B) S → A → C → G, cost 9

(C) S → B → D → G, cost 10

(D) S → A → D → G, cost 10

Answer: (B) S → A → C → G, cost 9

Explanation: f(A)=2+5=7, f(B)=3+4=7; f(C)=6+2=8 is lowest next; G via C has f=9+0=9, while every alternative route to G costs 10. Since h is admissible (h ≤ true cost at every node), A* returns the optimal path S→A→C→G with cost 9.

Q62 *Fill in the Blank · Easy*

In A* search, the evaluation function is f(n) = g(n) + _____.

Answer: h(n) (also accepted: h, hn, heuristic, h of n, heuristic function)

Explanation: g(n) = cost so far from the start; h(n) = estimated cost from n to the goal.

Q63 *Scenario-based · Medium [RW]*

A navigation app runs A* using straight-line (as-the-crow-flies) distance to the destination as h(n). Why is this heuristic admissible?

(A) Because roads are always straight

(B) No road route can be shorter than the straight-line distance

(C) Because it is computed with GPS

(D) Because it always equals the road distance

Answer: (B) No road route can be shorter than the straight-line distance

Explanation: The straight line is the shortest possible distance between two points, so it never overestimates the true road distance — that is admissibility.

20. Constraint Satisfaction Problems

Q64 *MCQ · Medium*

Australia map colouring (WA, NT, SA, Q, NSW, V, T) with 3 colours. At the very start, with all domains equal, which variable does the DEGREE heuristic choose first?

(A) WA (Western Australia)

(B) SA (South Australia)

(C) T (Tasmania)

(D) NSW (New South Wales)

Answer: (B) SA (South Australia)

Explanation: SA borders 5 regions (WA, NT, Q, NSW, V) — the most constraints on other unassigned variables. MRV can't break the tie at the start, so degree picks SA.

Q65 *Fill in the Blank · Medium [S]*

Crypt-arithmetic: SEND + MORE = MONEY, each letter a distinct digit and no leading zeros. The letter M must equal _____.

Answer: 1 (also accepted: one)

Explanation: Adding two 4-digit numbers gives at most 19,998, so the carry into the 5th digit (M) can only be 1. Full solution: 9567 + 1085 = 10652.

Q66 *Scenario-based · Easy [RW]*

Christ University models end-semester exam timetabling as a CSP (variables = exams, domains = time slots). Which of the following is a CONSTRAINT?

(A) Each exam is a variable

(B) The available time slots

(C) No student can have two exams in the same slot

(D) The list of courses offered

Answer: (C) No student can have two exams in the same slot

Explanation: Variables are exams, domains are slots, and constraints restrict which combinations of assignments are allowed — e.g. exams sharing a student must get different slots.

Q67 *Assertion-Reason · Medium*

Assertion (A): Forward checking can detect a dead end earlier than plain backtracking.

Reason (R): After each assignment, forward checking deletes inconsistent values from the domains of unassigned neighbouring variables.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: If any neighbour's domain becomes empty, forward checking backtracks immediately — before plain backtracking would have reached that variable.

Q68 *True/False · Easy*

The Minimum Remaining Values (MRV) heuristic selects the variable with the MOST legal values left.

Answer: False

Explanation: MRV ('fail-first') picks the variable with the FEWEST legal values, so failures are found early and the tree is pruned.

21. Means-End Analysis

Q69 *MCQ · Easy*

Means-End Analysis was the central technique of which early AI system?

(A) ELIZA by Weizenbaum

(B) General Problem Solver (GPS) by Newell & Simon

(C) MYCIN by Shortliffe

(D) Deep Blue by IBM

Answer: (B) General Problem Solver (GPS) by Newell & Simon

Explanation: GPS (Newell, Shaw & Simon, 1957–59) solved problems by repeatedly reducing the difference between the current and goal states.

Q70 *Scenario-based · Medium [RW]*

A warehouse robot must PUSH a desk to another room, but the desk has a lamp on it and PUSH requires the desk top to be clear. What does Means-End Analysis do next?

(A) Abandons PUSH and picks a random operator

(B) Sets up a sub-goal to satisfy PUSH's precondition (remove the lamp), then resumes

(C) Restarts the search from the initial state

(D) Applies PUSH anyway and ignores the precondition

Answer: (B) Sets up a sub-goal to satisfy PUSH's precondition (remove the lamp), then resumes

Explanation: This is operator sub-goaling: when the chosen operator's preconditions aren't met, MEA recursively solves the problem of reaching a state where they are, then applies the operator and continues toward the goal.

Q71 *Assertion-Reason · Hard*

Assertion (A): Means-End Analysis mixes forward and backward reasoning.

Reason (R): It first reduces the most important difference with a chosen operator, then solves the sub-problems of reaching that operator's preconditions and of getting from its result to the goal.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is NOT the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A

Explanation: Choosing an operator for the biggest difference and then working on its preconditions is backward-style; moving from the operator's result to the goal is forward-style. R describes exactly that mix.

Q72 *MCQ · Medium*

In Means-End Analysis, what does a 'difference table' record?

(A) The heuristic value of each state

(B) Which operators can reduce each kind of difference between current and goal states

(C) The cost of every path explored so far

(D) The order in which nodes were expanded

Answer: (B) Which operators can reduce each kind of difference between current and goal states

Explanation: The difference table indexes operators by the differences they reduce, so MEA can look up a relevant operator for the most important difference.